

Tuning Ferrimagnetic Heuslers for MRAM: Materials Driven Gains in Mn₃Ge MTJ Switching and TMR

Panagiotis Ch. Filippou^{1*}, Ikhtiar², See-Hun Yang¹, Sergey Faleev¹, Mahesh G. Samant¹, Jaewoo Jeong²

¹IBM Research–Almaden, San Jose, CA 95120, USA

²Samsung Semiconductor Inc., San Jose, CA 95134, USA

*email: panos.filippou@ibm.com

We propose a paradigm shift in materials for MRAM, enabled by ordered alloys and ferrimagnetic Heusler alloys, which decouple magnetic anisotropy, of magnetocrystalline in origin, from magnetic moment. This decoupling allows simultaneous realization of high thermal stability ($E_b > 60$ kT) and ultrafast switching (< 2 ns), overcoming a long-standing bottleneck in spin-transfer torque (STT) MRAM design. We further identify key materials and fabrication challenges, including interface oxidation, interdiffusion, and epitaxial growth. This work establishes ordered alloy MTJs as a path toward next-generation, cache-class, ultra-fast nonvolatile memory.

Index Terms— Chemical Templating Layers, Ferrimagnet, Heusler Alloys, MRAM

I. INTRODUCTION

ORDERED ALLOY BASED FREE LAYERS are promising candidates for ultrafast switching with a low write error rate (WER) floor, while simultaneously enabling low switching currents and advanced architectures for highly scalable, next-generation STT-MRAM [1,2].

Perpendicular magnetic anisotropy (PMA) in magnetic tunnel junctions (MTJs) has traditionally relied on interfacial anisotropy [3], particularly at the MgO/CoFeB interface, which scales unfavorably as device dimensions approach the sub-20 nm regime. The reduction in magnetic volume limits the achievable thermal stability (even in double-interface MgO/CoFeB structures, while ultrathin free layers can also suffer from increased damping and interfacial variability) making it increasingly difficult to simultaneously achieve high thermal stability, low switching current, and low write-error rate.

As an alternative pathway, engineering the aspect ratio of the free layer to introduce perpendicular shape anisotropy (PSA) [4,5], in conjunction with interfacial anisotropy, enables large effective anisotropy and thus high thermal stability even at deeply scaled dimensions. PSA-based MTJs and multilayer free-layer designs have shown very high thermal stability factors, with demonstrated scalability down to ~ 2 –3 nm nodes. While such approaches enable sufficient thermal stability, the switching characteristics are strongly governed by the energy barrier E_b , which sets the thermal stability factor Δ and directly influences both the write error rate and the critical switching current. This highlights the necessity of achieving an optimal balance for reliable, low-error, and fast switching operation.

When considering deep scaling of MTJs, the saturation magnetization (M_s) of the free layer emerges as a critical parameter. From the perspective of angular momentum conservation, the spin-transfer torque required to switch the magnetization must supply the angular momentum of the entire free layer volume, which scales with M_s . In this context, low M_s free layers are particularly advantageous. Additionally, ordered alloys whose PMA is derived from their crystal structure (i.e.

magnetocrystalline anisotropy) can exhibit high values of PMA. Therefore, materials with inherently low or tunable M_s , such as ordered alloys and their engineered multilayers can achieve ultrafast ($< ns$) switching, high thermal stability while maintaining an acceptable WER floor at deeply scaled nodes.

II. MTJ DEVICE PERFORMANCE LINKED TO ORDERED ALLOY

In free layers comprised of Mn₃Ge ordered alloys, PMA arises from bulk crystal symmetry rather than interfacial effects [6], while the net magnetization is intrinsically reduced due to its antiferromagnetically aligned sublattices. As a result, thermal stability can remain high even as M_s is reduced.

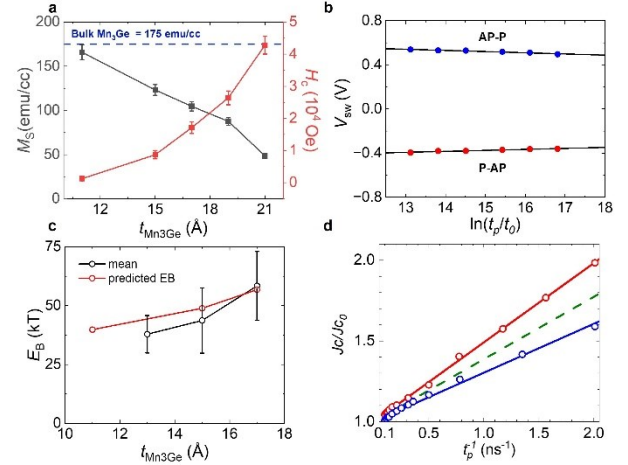


Fig. 1. a) Magnetization of Mn₃Ge vs thickness. b) and c) switching of Mn₃Ge free layer in MTJ and thermal stability d) sub-10ns switching of Mn₃Ge

This decoupling enables aggressive scaling and positions ordered alloys as uniquely suited for ultra-fast, high-density MRAM. This material's advantage translates directly into device-level performance. MTJs incorporating ordered alloy free layers such as Mn₃Ge have demonstrated simultaneous high retention energy barriers and nanosecond-scale switching.

Fig. 1 shows the M_s of Mn₃Ge which is more than 6x times lower than of CoFeB, along with stt-switching voltage in MTJs made of Mn₃Ge free layer and with an extracted E_b that can be

tuned to be high based on the thickness of the Mn_3Ge layer here. Lastly ultrafast stt-switching of this Mn_3Ge free layer shows that sub-10ns switching, down to 0.5ns requires a surplus current application which is estimated to be $\sim 38\%$ increase whereas a free layer made of CoFeB requires $\sim 200\%$ current increase [1]. Furthermore, low magnetization reduces spin-transfer torque switching current density while maintaining strong anisotropy fields, leading to efficient and reliable operation with steep WER curves and reduced overdrive requirements, as seen in Fig. 2.

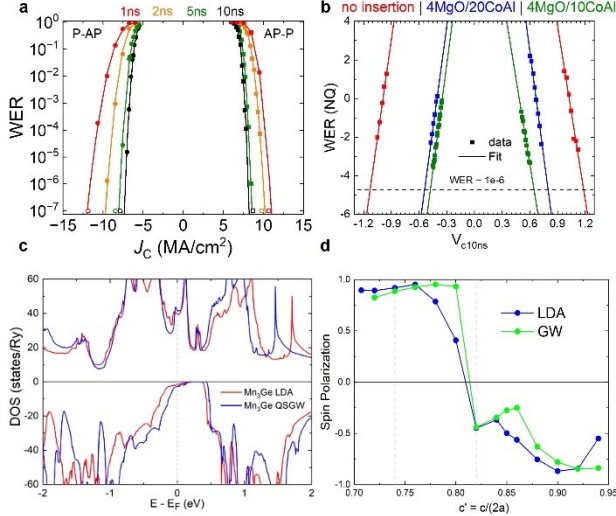


Fig. 2. a) WER curves for Mn_3Ge free layer based MTJ. b) improved STT performance via chemical templating layer engineering. c) Mn_3Ge DOS at 4.03Å in-plane lattice parameter d) Spin Polarization calculation of Mn_3Ge vs degree of tetragonality of the Mn_3Ge .

These improvements persist even at device dimensions below 20 nm, overcoming the scaling limitations inherent in interfacial-anisotropy-based CoFeB systems. Notably, multilayer free-layer architectures, combining ordered alloys with high-spin-polarization layers or Heusler compounds may enable simultaneous optimization of PMA and tunnel magnetoresistance (TMR), further enhancing readout fidelity without sacrificing switching performance. Beyond single-layer materials, engineered free-layer architectures provide an additional degree of freedom. Coupled or exchange-coupled composite free layers, consisting of multiple layers or ordered alloy sublayers separated by lattice-matched spacer layers, allow independent tuning of switching dynamics and retention as has been seen by utilizing RuAl or IrAl spacer layers [7]. Engineering the chemical templating layers, with the incorporation of resistive layers as MgO , ScN also lead to enhanced device performance from the Mn_3Ge free-layer MTJ, shown in Fig. 2. Lastly, tuning the ordered alloy lattice parameter, e.g. via the appropriate selection of chemical templating layer, can yield higher Spin Polarization as seen from the calculation in Fig. 2 [1]. These approaches establish a versatile design framework in which device performance can be engineered through ordered alloy layer composition, strain, thickness, and coupling and create a regime where low switching energy and high retention coexist.

Despite these advantages, several materials and fabrication challenges remain. Ordered alloys are highly sensitive to interfacial chemistry, particularly oxidation during MgO barrier deposition and interdiffusion at elevated processing temperatures. However, targeted solutions such as high-angle deposition geometries, diffusion barrier layers, and optimized annealing protocols have already demonstrated substantial mitigation of these issues [8]. Continued advances in chemical templating layers, epitaxial growth, and interface engineering are expected to further unlock the full potential of ordered alloy MTJs.

III. CONCLUSION

MRAM has now had a materials physics breakthrough with ferrimagnetic Heuslers, exhibiting, low M_s , high bulk PMA, tunable magnetic structure, all of which enable low switching energies and high thermal stability. Ordered alloy and ferrimagnetic Heusler materials redefine the design space of spintronic devices by decoupling switching energy from thermal stability, enabling ultra-fast, low-power, and highly scalable MRAM technologies. Through a combination of materials innovation, interface engineering, and multilayer device architectures, these systems overcome long-standing tradeoffs that have limited conventional approaches. While challenges remain in materials integration and process control, recent progress demonstrates clear pathways to resolution. The convergence of these advances; positions ordered alloy MTJs as a leading candidate for next-generation memory, with the potential to achieve cache-level performance at dramatically reduced energy consumption.

REFERENCES

- [1] Garg, C. *et al.*, “Ferrimagnetic Heusler tunnel junctions with fast spin-transfer torque switching enabled by low magnetization” *Nat. Nanotechnol.* 1–6 (2025) doi:10.1038/s41565-024-01827-7.
- [2] Gottwald, M. G. *et al.* “First Demonstration of High Retention Energy Barriers and 2 ns Switching, Using Magnetic Ordered-Alloy-Based STT MRAM Devices” *VLSI Technology and Circuits* 1–2 (2024). doi:10.1109/VLSITechnologyandCir46783.2024.10631319.
- [3] H. Sato *et al.*, “Properties of magnetic tunnel junctions with a $\text{MgO}/\text{CoFeB}/\text{Ta}/\text{CoFeB}/\text{MgO}$ recording structure down to junction diameter of 11 nm,” *Appl. Phys. Lett.*, vol. 105, no. 6, p. 062403, (2014).
- [4] Perrissin, N. *et al.* “A highly thermally stable sub-20 nm magnetic random-access memory based on perpendicular shape anisotropy” *Nanoscale* 10, 12187–12195 (2018).
- [5] Jinnai, B. *et al.* “High-Performance Shape-Anisotropy Magnetic Tunnel Junctions down to 2.3 nm” *IEDM* 24.6.1-24.6.4 (2020). doi:10.1109/IEDM13553.2020.9371972.
- [6] P. C. Filippou *et al.*, “Chiral domain wall motion in unit-cell thick perpendicularly magnetized Heusler films prepared by chemical templating,” *Nature Communications*, vol. 9, no. 1, p. 4653, (2018).
- [7] P. Ch. Filippou *et al.*, “Heusler-based synthetic antiferromagnets,” *Science Advances*, vol. 8, no. 8, p. eabg2469, (2022).
- [8] Jeong, J. *et al.* “Enhancement in tunnel magnetoresistance for magnetic tunnel junctions in Mn based Heusler compounds” *Appl. Phys. Lett.* 126, 252402 (2025).